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Bayesian Techniques in Macroeconomics

Université Paris Dauphine – PSL
Master in Quantitative Economics, second year

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Chapter 4

Bayesian Estimation of DSGE Models

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Handout Objectives

- Estimating a DSGE model using Bayesian techniques.
- Decomposing business cycles into multiple sources of exogenous disturbances.
- Producing and evaluating forecasts.

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Session roadmap

Economic intuition. Priors encode disciplined information about structural mechanisms. The likelihood confronts them with data. **Bayesian intuition.** The posterior combines both sources and MCMC quantifies uncertainty beyond the mode. **Model and data.** Check support, measurement equations, identification, and independent shocks. **MATLAB/Dynare exercise.** Estimate, inspect acceptance and mixing, and report posterior IRFs with credible intervals. **Takeaway.** Posterior claims require reproducible convergence and sensitivity diagnostics.

1. A Bayesian perspective

We estimate a state-space linear model using Bayesian techniques as in Fernandez-Villaverde and Rubio-Ramirez [2004] and Smets and Wouters [2007]. Non-calibrated parameters are estimated using the Bayesian posterior inference (see An and Schorfheide [2007] for an extensive presentation). As exemplified by Smets and Wouters [2003] and Smets and Wouters [2007], this approach can fruitfully be applied to macroeconomic models as it allows a DSGE model to perform as well as an unrestricted VAR model in explaining macroeconomic time series. In our estimation we follow closely each of the steps defined by Smets and Wouters [2003].[2]

1.1 Building the likelihood function

Recall that in the first handout, any macroeconomic model with expectations can be cast in a compact form as follows:

$$\mathbb{E}_t \left[f_\Theta \left(x_{t+1}, x_t, x_{t-1}, \epsilon_t \right) \right] = 0 \tag{1}$$

where f_Θ are the non-linear relations across variables x_t, stochastic innovations ϵ_t, for a given set of structural parameters Θ.

By applying linearization methods on the system of nonlinear equations, it is possible to get a solution of the form

$$\hat{x}_t = F\hat{x}_{t-1} + G\epsilon_t, \tag{2}$$

where elements in matrices F and G are functions of structural parameters Θ.

Let H be the selection matrix of observable variables – a subset of $\hat{x}_t$ – such that the observable vector is $y_t = H\hat{x}_t + v_t$ with v_t the measurement errors. Then it is possible following the application of a filter (e.g. Kalman, inversion, particle,...) to get the errors of prediction:

$$S_t = Y_t - E_{t-1}\{y_t\},$$

[2]The successive steps of this estimation strategy are presented by Smets and Wouters [2003] as follows: "this econometric approach attempts to provide a full characterisation of the observed data series. For example, following Sargent [1989], a number of authors have estimated the structural parameters of DSGE models using classical maximum likelihood methods. These maximum likelihood methods usually consist of four steps. In the first step, the linear rational expectations model is solved for the reduced form state equation in its predetermined variables. In the second step, the model is written in its state space form. This involves augmenting the state equation in the predetermined variables with an observation equation which links the predetermined state variables to observable variables. In this step, the researcher also needs to take a stand on the form of the measurement error that enters the observation equations. The third step consists of using the Kalman filter to form the likelihood function. In the final step, the parameters are estimated by maximising the likelihood function. Alternatively within this strong interpretation, a Bayesian approach can be followed by combining the likelihood function with prior distributions for the parameters of the model, to form the posterior density function. This posterior can then be optimised with respect to the model parameters either directly or through Monte-Carlo Markov-Chain (MCMC) sampling methods."

where $E_{t-1}\{y_t\}$ is an output from the statistical filter and Y_t is a sample with n observed variables and T periods.

The prediction errors of the estimated model follow $S_t \sim \mathcal{N}(0, \Omega_t)$ conditionally on past data, with Ω_t the variance-covariance of prediction errors, the general expression of the likelihood function reads as follows:

$$\log \mathcal{L}(\theta|Y_T, ...Y_1) = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_{1}^{T}\log(|\Omega_t|) - \frac{1}{2}\sum_{1}^{T} S_t'\Omega_t^{-1}S_t. \tag{3}$$

The aim is to estimate the vector of parameters θ, a subset of all the structural parameter vector Θ. Here θ is a subvector. Finally, in most software, it is easier to minimize than maximize, so the maximum-likelihood estimator reads as follows:

$$\hat{\theta}_{MLE} = \arg\min_{\{\theta\}} \{-\log \mathcal{L}(\theta|Y_T, ...Y_1)\} \tag{4}$$

Note that prediction errors S_t are actually a function of estimated parameters θ, i.e. some parameter values may improve the forecasting performance of the model, and thus reduce the volatility of the errors of predictions Ω_t. So the idea is to pick a value of $\hat{\theta}_{MLE}$ that maximizes the joint density.

1.2 Adding prior information

The Bayesian approach consists of defining an *a priori* distribution $p(\theta)$ over the structural parameters θ, including both a prior value and a distribution. The posterior combines prior information with the likelihood. A posterior summary is not generally a simple average of the prior mean and the MLE. The choice of prior hyperparameters requires an intuition for their role in the economy. We follow the standard practice of the literature in choosing one of the following probability distributions for parameters: the Normal distribution ($\mathcal{N}$), the Beta distribution ($\mathcal{B}$, for parameters with values lying between 0 and 1), the Gamma distribution ($\mathcal{G}$, for priors which allow relatively diffuse and positive support) and the Inverse Gamma distribution ($\mathcal{IG}$, for shock standard deviations, whose support must be positive).

The estimates of the parameters are updated using the dataset Y according to Bayes' rule:

$$p(\theta|Y) = \frac{p(Y|\theta)p(\theta)}{p(Y)} \propto L(\theta|Y)p(\theta), \tag{5}$$

where $L(\theta|Y)$ denotes the likelihood function, $p(\theta|Y)$ is the posterior distribution of parameters and $p(Y)$ is the marginal likelihood (marginal data density).

Obtaining a closed-form normalized posterior is generally not possible. However, the posterior distribution of the model parameters can be constructed numerically by applying the Markov Chain Monte Carlo (MCMC) methods. Finally, we compute the posterior moments of the parameters using a sufficiently large number of draws, having made sure afterwards the MCMC algorithm has converged.

$$\hat{\theta}_{BE}^{\text{mode}} = \arg\max_{\theta} L(\theta|Y)p(\theta) \tag{6}$$

One main advantage of DSGE models compared to VARs concerns the evaluation of monetary policy. DSGE models replicate time series by the responses of agents to different sequences of shocks. Once we have estimated all the sequences of shocks and the reaction parameters of households, firms and monetary authorities, we can perform counterfactual analyses by changing the parameters of monetary policy. Counterfactual scenarios can be derived from the estimated model by optimizing the policy coefficients in the Taylor rule to see how well the economy would have performed under an alternative policy regime.

1.3 An illustration with the RBC filter

Generating artificial series. Consider our 3 equation RBC model estimated with maximum likelihood and inversion filter (we could have done the same with the Kalman filter, but it does not fit into one single MATLAB file).

Recall that in the previous handout we had done the following:

- Simulating a model for 200 periods, we next assumed that only consumption was observable, and so we used consumption to retrieve the sequence of productivity shocks that generated the data.
- The distribution of productivity errors is given by $\varepsilon_t \sim \mathcal{N}(0, Q)$, let us assume that we want to estimate Q, i.e. the variance(-covariance) of errors. In this separate illustration, $Q = 1$ is a numerical scale for inference, not a realistic calibration of technology shocks.
- The likelihood function is given by

$$\log L(Q|\varepsilon_T, ...\varepsilon_1) = -\frac{nT}{2}\log(2\pi) - \frac{1}{2}\sum_1^T \log(|Q|) - \frac{1}{2}\sum_1^T \varepsilon_t' Q^{-1}\varepsilon_t.$$

- The MLE minimizes $-\log L$. The inversion Jacobian is omitted here only because F, G, H are fixed and only Q varies.

Setting prior information. Now we can introduce Bayesian econometrics. Which prior can we set for Q? The best distribution would be one with positive support, but for clarity we choose the simplest distribution: a Gaussian density restricted to $Q > 0$. The normalizing constant of this truncation does not affect optimization or MH acceptance ratios. Dynare, and most macroeconomic papers, express any distribution into two moments: the mean and standard deviation of the distribution. Since a Gaussian distribution is already expressed in terms of its mean and standard deviation, it is straightforward to parameterize the prior.

Let us assume that the econometrician has some information about the distribution of Q and specifies the prior kernel $Q \sim \mathcal{N}(1, 0.10^2)$. The log of the joint probability density of observing our data (the log-likelihood function) is now augmented by the probability density of some estimated parameters, that we refer to as priors, denoted $P(\theta)$. Since our prior is $Q \sim \mathcal{N}(1, 0.10^2)$, the log-posterior kernel now reads as follows:

$$\underbrace{\log L(Q|\varepsilon_T, ...\varepsilon_1)}_{\text{likelihood}} + \underbrace{\log p(Q, 1, 0.10)}_{\text{priors}}$$

where $p(.)$ is a Gaussian density with mean 1 and standard deviation 0.10. This translates into MATLAB as:

```
mu_Sige = 1;
sd_Sige = .1;
lnprior = @(x)  -0.5*((x-mu_Sige)/sd_Sige).^2 - log(sd_Sige*sqrt(2*pi));
```

Let us now explore the topology of the prior function in Figure 1. We can see that $p(Q, 1, 0.10)$ is higher when Q gets closer to the prior mean 1, so the prior information curbs the likelihood function in order to take some knowledge from the econometrician about the set of estimated parameters θ.

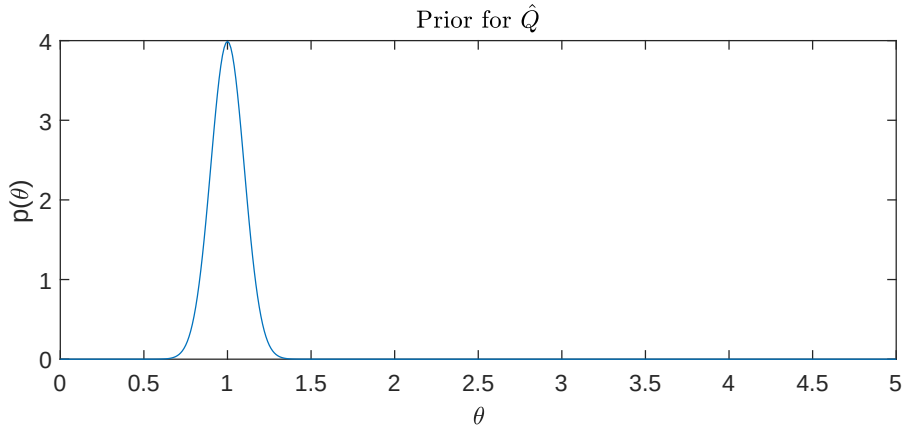


Figure 1: Prior kernel for the innovation variance Q

Optimizing the posterior density. Like for maximum likelihood, we must optimize an objective function. Now this objective function is the posterior density: the latter is proportional to the product of the prior density and the density of the sample (i.e. the likelihood function). This posterior distribution takes into account both the contribution of the data (i.e. the log-likelihood) and the prior information. We solve numerically the following minimization problem to obtain the mode of the posterior distribution:

$$\hat{Q}_{BE}^{\text{mode}} = \arg\min_{Q} -\log L(Q|Y) - \log p(Q) \tag{7}$$

```
theta0 = [0.01];
theta_MLE = fmincon(@(x) -llk(x),theta0,[],[],[],[],1e-8);
theta_BE = fmincon(@(x) -(llk(x) + lnprior(x)),theta0,[],[],[],[],1e-8);
disp('Mode:')
disp(['MLE: Est. Q ' num2str(theta_MLE)])
disp(['BE : Est. Q ' num2str(theta_BE)])
```

A posterior mode is a maximizer of the posterior density. Having obtained the mode, we also need the mean of the density as well as the dispersion of the density. Numerical quadrature is feasible in this scalar example. MCMC illustrates a method that also extends to a higher-dimensional DSGE posterior. Economists typically use Markov Chain Monte Carlo (MCMC) methods – in particular the Metropolis–Hastings algorithm – for this purpose.

Drawing uncertainty through Monte Carlo methods. The main purpose of the MH algorithm is to explore the parameter space of Q by weighting the outcomes appropriately. A Markov chain is a stochastic model describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event.

1. **Get covariance of estimated parameters.** We first need the local approximation to the posterior covariance of θ. There are different ways to obtain it. In this example, we obtain it from a local Laplace approximation at the posterior mode:

$$\Sigma_\theta \simeq \left[- \left. \frac{\partial^2 [\log L(\theta|Y) + \log p(\theta)]}{\partial\theta \, \partial\theta'} \right|_{\theta=\hat{\theta}_{BE}^{\text{mode}}} \right]^{-1}, \tag{8}$$

2. **Markov Chain initialization.** We initialize each Markov Chain at a valid parameter vector, here the estimated posterior mode $\theta_0 = \hat{\theta}_{BE}^{\text{mode}}$. Next, the chain evolves from the last retained state, not always from the mode. Use dispersed admissible initial states when comparing chains.

3. **Draw a new candidate.** We next draw for each $i = \{1, ...N\}$ a new candidate θ_i^{cand} based on a random Gaussian perturbation of covariance $c^2\Sigma_\theta$:

$$\theta_i^{cand} \sim \mathcal{N}(\theta_{i-1}, c^2\Sigma_\theta) \tag{9}$$

   Note that c is a scale parameter. It critically determines how large is the exploration space. If c is "too" small, the explored parameter space will be too narrow in the neighborhood of $\hat{\theta}_{BE}^{\text{mode}}$. Conversely, if the variance of perturbations is too large, many proposals have low posterior density and are rejected. The chain can then remain at the same state for many draws. I discuss how to pick a not "too-big-too-small" c value, using a criterion called the acceptance rate that I discuss hereafter.

4. **Compute the acceptance probability of the candidate.** Is this new candidate θ_i^{cand} more favored by the data than the previous retained state θ_{i-1}? For this symmetric random-walk proposal, proposal densities cancel. The acceptance probability is:

$$\alpha_i = \min\left(\frac{L(\theta_i^{cand}|Y) \times p(\theta_i^{cand})}{L(\theta_{i-1}|Y) \times p(\theta_{i-1})}, 1\right). \tag{10}$$

   A higher-density candidate is accepted with probability one. For an asymmetric proposal, include the reverse-to-forward proposal-density ratio. Numerically compare $\log u_i$ with $\min\{0, g(\theta_i^{cand}) - g(\theta_{i-1})\}$, where $g = \log L + \log p$.

5. **Accept or reject the candidate.** Note that we are not trying to find the best value of θ (we already have it, it's the mode of the posterior distribution), so the decision to accept or reject must also include a noise in order to accept candidates with low probability.

$$\theta_i = \begin{cases} \theta_i^{cand} \text{ for } \mathcal{U}(0,1) < \alpha_i \text{ (accept)} \\ \theta_{i-1} \qquad\qquad\qquad\quad \text{(reject otherwise)} \end{cases}$$

   Store the retained state after every step, including rejections. Under suitable ergodicity conditions, the empirical distribution approaches the posterior. Keeping only accepted moves would distort posterior probabilities.

6. **Repeat N times step 3, 4 and 5.**

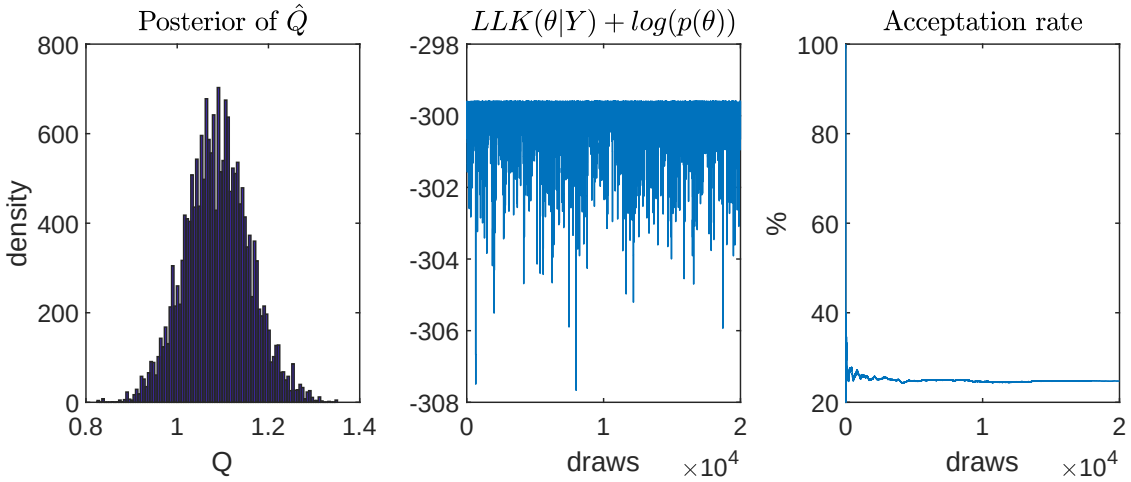
7. **Truncate initial values.** Truncate a fraction of the first draws (to start to the nth draw) to reduce sensitivity to initialization. Burn-in alone does not establish convergence. The Markov chain now reads as $\{\theta_i\}_{i=n}^N$.

8. **Analysis.** Analyse the posterior distribution using the histogram of $\{\theta_i\}_{i=n}^N$. Compare the prior with its posterior distribution. Check the empirical acceptance fraction, trace plots, autocorrelation and effective sample size. For multivariate random-walk proposals, an acceptance rate near 20%–35% is a useful tuning heuristic, not a convergence test or a universal target.

   (a) A low acceptance rate often means that proposal steps are too large. Reduce c and check whether mixing improves.

(b) A very high acceptance rate often means that proposal steps are too small. Increase c cautiously and assess mixing, not acceptance alone. Reject proposals outside the parameter support rather than clipping them. Use dispersed starting points when comparing multiple chains.

MATLAB box 1 bayesianRBC.m



Shape	Dynare inputs	Support	Example(s)
normal_pdf	μ, σ	$(-\infty, +\infty)$	Policy parameters, utility curvatures,
gamma_pdf	μ, σ	$(0, +\infty)$	Elasticities, utility curvatures, trends
beta_pdf	μ, σ	$[0, 1]$	Probabilities, AR-MA terms, shares...
inv_gamma_pdf	μ, σ	$(0, +\infty)$	Standard deviations
uniform_pdf	μ, σ	$[a, b]$	Bounded parameters, with finite $a < b$
weibull_pdf	μ, σ	$[0, +\infty)$	For standard deviations, in particular of measurement shocks
inv_gamma2_pdf	μ, σ	$(0, +\infty)$	Standard deviations

Table 1: Prior distributions available in Dynare

2. Bayesian estimation with Dynare

Dynare also includes a quick and easy way to estimate linear DSGE models with the Kalman filter and prior information. Most of the code is going to be very similar to maximum likelihood estimation from the previous handout. The only difference lies in the specification of priors, and how from the maximization of the posterior density, we obtain the posterior distribution.

2.1 Priors

When we are estimating with Bayesian techniques, we need to specify prior distributions for each parameter to be estimated. In the estimated params section, for each estimated parameter, we declare a distribution, and specify parameters relevant for that distribution. The syntax for this is:

[parameter_name], [initial_value (optional)], [lower_bound (optional)], [upper_bound (optional)], [prior_shape],

[prior_mean], [prior_standard_deviation], [p_3 (optional)], [p_4 (optional)], [jscale (optional)].

Table 1 lists Dynare inputs: μ is a mean and σ a standard deviation, not generally native shape parameters. Mathematical normal notation remains $\mathcal{N}(\mu, \sigma^2)$. A beta prior requires $0 < \mu < 1$ and $0 < \sigma^2 < \mu(1-\mu)$. Prior specification (in terms of shape, mean and standard deviation) allows to give more information about the distribution of the parameters and to avoid some regions where the likelihood function could be flat.

To illustrate the guideline for choosing a prior, let us consider first the AR(1) term of the productivity shock process ρ^A. To have a stationary AR(1), $|\rho^A| < 1$ is required. Here we additionally impose positive persistence, $0 < \rho^A < 1$, to obtain a non-oscillating response. The beta distribution meets this requirement as it has a support on $[0, 1]$ and will not allow this parameter to be above 1 or below 0. We therefore assume $\rho^A \sim \mathcal{B}(0.5, 0.2)$. In estimated_params; block, we can add the following prior distribution as follows:

```
estimated_params;
rho_A, .8, 0, 1, beta_pdf, .5, .2;
end;
```

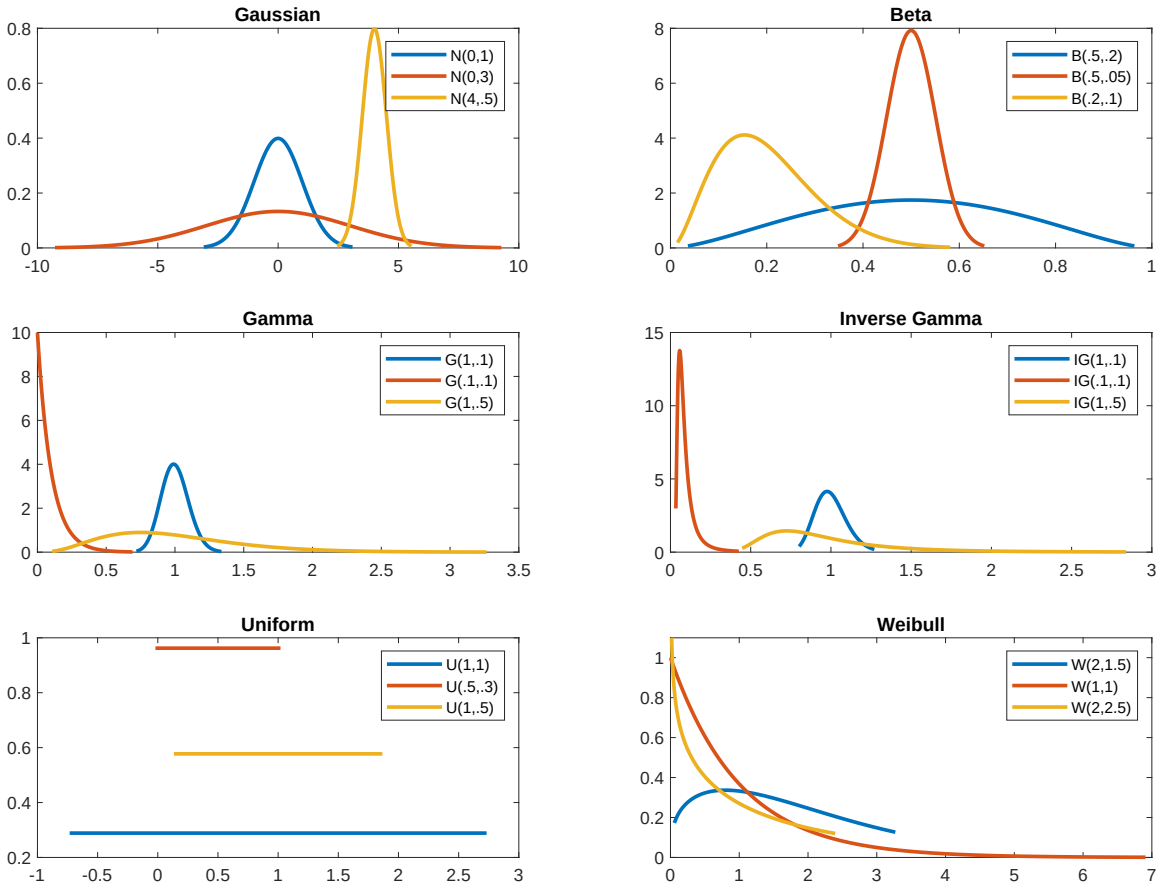


Figure 2: Prior distributions

In the previous specification, we assume that each parameter is likely to have a mean of 0.5 and a standard deviation of 0.20. This prior is more dispersed than the next example, but it is not uninformative. Its shape weights values unequally. Conversely, an informative prior with a mean of 0.90 and a standard deviation of 0.01 would keep the estimate close to 0.90. Economists choose how informative their priors should be. For instance, if monetary policy shocks are known not to be strongly autocorrelated, a prior mean of 0.20 with a standard deviation of 0.05 could be fairly acceptable. Neither diffuse nor tight priors are automatically preferable. Justify support and dispersion and report sensitivity.

Let us consider another example. The estimated standard deviation of a shock has to be positive, this constraint excludes *de facto* the Normal distribution as it has a negative support. A Beta distribution could be motivated as we want estimated shocks to be small otherwise our first order approximation becomes irrelevant with large shocks. However, demand and supply shocks may have a standard deviation higher than 1, which excludes the Beta distribution. It is then possible to use a Uniform distribution, as well as a Gamma or Inverse Gamma distributions. Here, we follow Smets and Wouters [2007] using an Inverse Gamma distribution with a mean of 0.10 and a standard deviation of 2 such that $\sigma^A \sim \mathcal{IG}(0.1, 2)$. Accordingly, we add in the Dynare file the lines which code the estimation of the shocks' volatility.

```
stderr
↪  e_a,        .7,        0,        inf,        inv_gamma_pdf,        .1,        2;
```

2.2 Estimation and posteriors

Once the priors are written in Dynare, we have to declare which variables are observable (and they should appear in the data file as well). Here, we estimate using one time series:

```
varobs        y;
```

Finally, we declare the estimation command as we did for maximum likelihood estimations. There are lots of options for Bayesian estimation which can go here. Consult the Dynare Manual for an overview of these options, and resources on Bayesian estimation for what they mean.

```
estimation(datafile=mydata,first_obs=1,
mh_replic=20000,mh_nblocks=2,mh_jscale=1.7,
prefilter=1,bayesian_irf) y c k a;
```

Here, mh_replic is the number of draws per chain and mh_nblocks is the number of chains. This example uses two chains of 20,000 draws. The proposal scale is 1.7. These settings are not a convergence guarantee. Compare the chains, inspect their traces and autocorrelation, and report effective sample size and convergence diagnostics. Longer chains are useful only if they explore the posterior adequately. The code uses stationary Kalman initialization for the stationary model. The alternative option lik_init=2 initializes the state covariance with a large diagonal matrix. It does not detrend non-stationary data. Option prefilter=1 removes the empirical mean but not a trend. The series must therefore be transformed consistently with the measurement equations before estimation. Finally, mh_jscale controls the proposal standard deviation. Decrease it when acceptance is too low and increase it cautiously when acceptance is very high. Always assess mixing and convergence rather than using acceptance as the only diagnostic.

The supplied Bayesian RBC file scales its productivity innovation by 1/100. Its shock standard deviation is consequently expressed in percentage points, whereas the observed log deviations are in decimal units. The following console extracts and figures are historical illustrations, not new Dynare 7 estimation results for the current downloaded sample.

Starting the estimation exercise, Dynare loads our data file, and starts to find the posterior mode using the csminwel algorithm. In the historical output below, the initial value of the log-posterior kernel is 540.6002, the algorithm employed by Dynare aims at maximising the value of the log-posterior kernel by adjusting the estimated parameters.

console response

```
Initial value of the log posterior (or likelihood): 540.6002
——————
f at the beginning of new iteration, -540.6001819383
Predicted improvement: 450.236373200
lambda = 1; f = -540.5959836
lambda = 0.33333; f = -761.5328209
lambda = 0.64439; f = -851.6090736
Norm of dx 0.30008
—-
```

At the final iteration of the algorithm (here 58 iterations), the improvement performed in the last iteration is low enough for the algorithm to stop the optimisation. The negative log-posterior kernel is now approximately -895.6. Dynare then prints in the console the mode of the posterior distribution:

console response

```
RESULTS FROM POSTERIOR ESTIMATION
parameters
 prior mean mode s.d. prior pstdev
rho_A 0.500 0.9842 0.0085 beta 0.2000
standard deviation of shocks
 prior mean mode s.d. prior pstdev
e_a 0.100 1.1003 0.0454 invg 2.0000
```

After finding the mode, the Metropolis-Hastings algorithm is employed. For each chain, it samples the number of draws we specified in the estimation() command.

Here, we have two chains of 20 000 draws. At the end of the sampling, we will get 40 000 samples. Dynare will automatically drop a specified fraction of draws (the percentage of drops can be manually set in the estimation command). Dynare computes uncertainty (credible interval size can be manually set in the estimation command) and the mean of each parameter using the sequence of random samples from Metropolis-Hastings output.

console response

```
Log data density is 889.521825.
parameters
 prior mean post. mean 90% HPD interval prior pstdev
rho_A 0.500 0.9817 0.9690 0.9946 beta 0.2000
standard deviation of shocks
 prior mean post. mean 90% HPD interval prior pstdev
e_a 0.100 1.1064 1.0335 1.1844 invg 2.0000
```

Dynare prints the mean and 90% highest posterior density interval upper and lower bounds as well

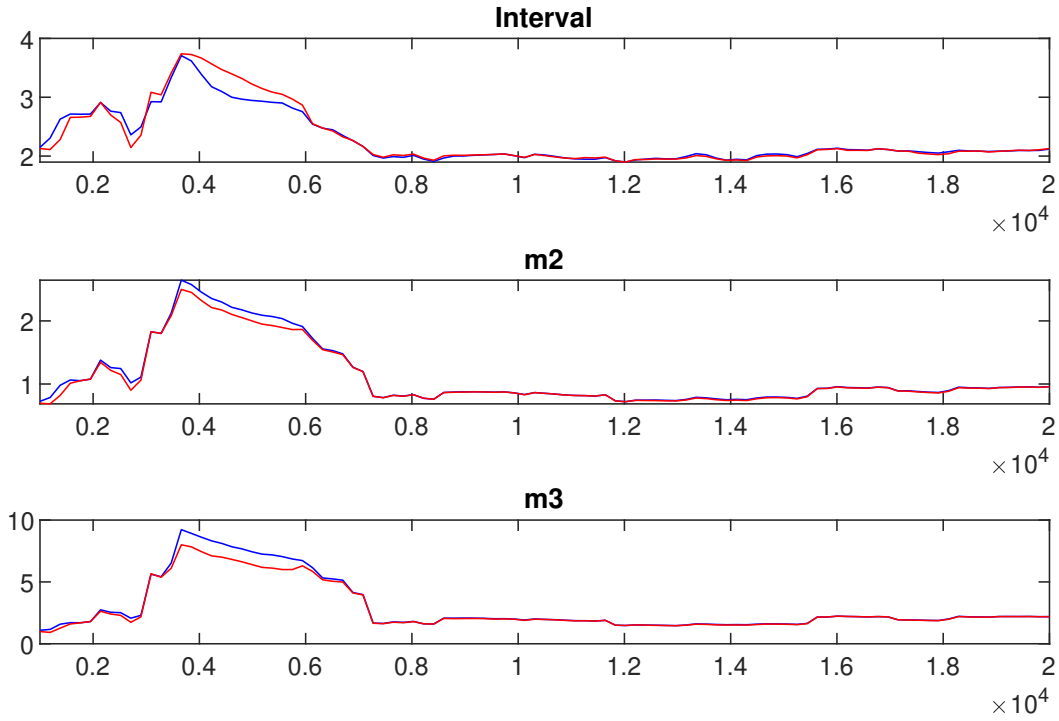


Figure 3: Univariate convergence diagnostic, Brooks and Gelman (1998)

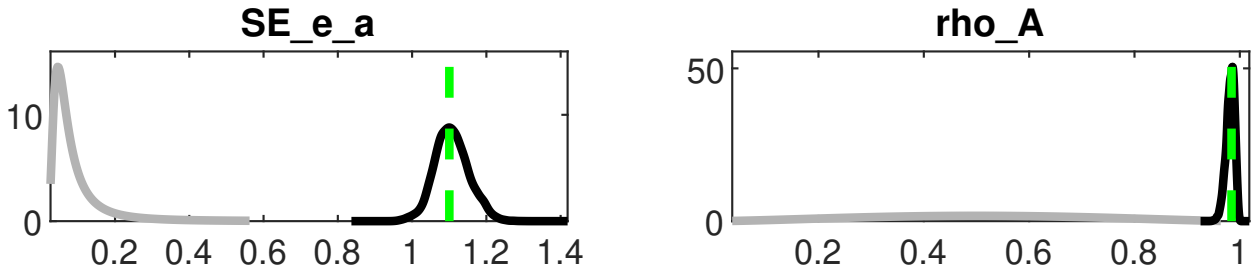


Figure 4: Priors (grey) and posteriors (black) distributions of estimated parameters and mean (green).

as the prior standard deviation of the parameter. The mean and mode answer different questions. The posterior mean minimizes posterior expected squared-error loss, while the mode locates the highest density. Dynare also runs some diagnostics to assess the convergence.

Figure 3 reports some of the diagnostics computed by Dynare to examine the convergence of the MCMC chains as in Brooks and Gelman [1998]. Here for instance, the red and blue lines should approach one another. If disagreement persists, check mixing, proposal scale and multimodality before increasing chain length. Do not change priors merely to improve a diagnostic.

Figure 4 compares the prior and posterior distributions. A posterior may remain centered near the prior mean while becoming much narrower, so a shift in the mean is not necessary for learning. Conversely, a shift alone does not establish identification. Compare dispersion, inspect the likelihood, check convergence, and assess sensitivity to reasonable priors. The acceptance rate must be diagnosed from the chains, not inferred from the prior–posterior plot.

Stored in	Description
oo_.posterior_mode	Parameters and shock standard errors at the posterior mode.
oo_.posterior_std	Standard error of the estimated values at the posterior mode.
oo_.posterior_density	Density of the posterior for each parameter (x,y values for each parameter).
oo_.prior_density	Density of the prior for each parameter (x,y values for each parameter).
oo_.SmoothedVariables	Estimated values of each endogenous variable for all time periods.
oo_.SmoothedShocks	Estimated values of each shock for all time periods.

Table 2: MATLAB's workspace after a Bayesian estimation

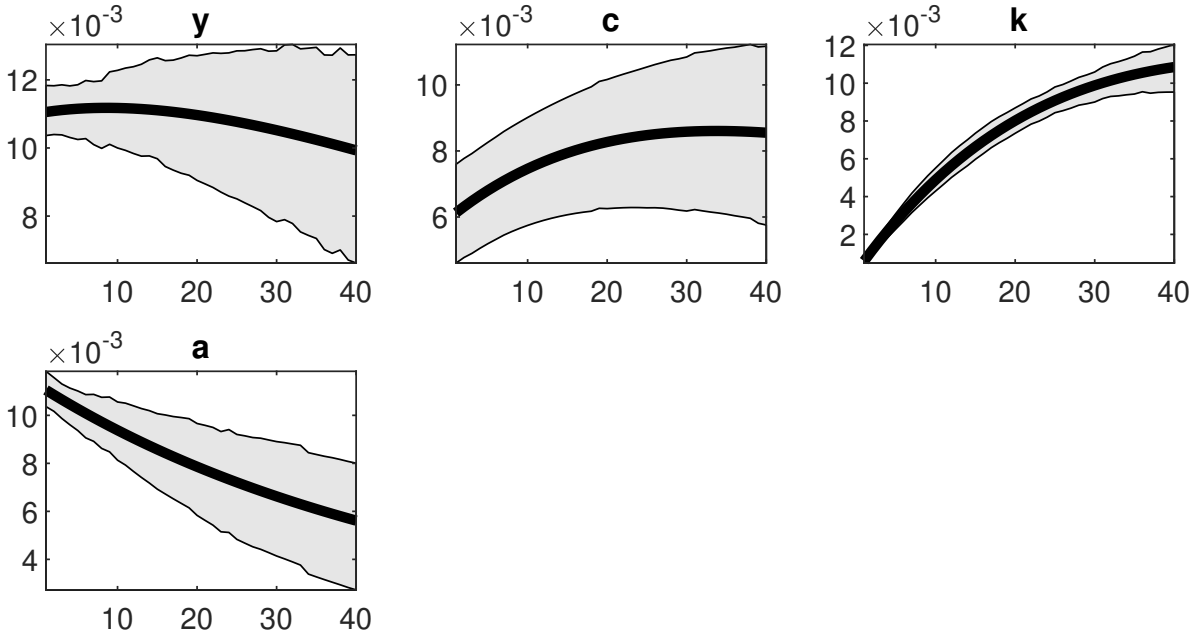


Figure 5: System response after an estimated supply shock

2.3 Bayesian system response

It is also worthwhile to have a look at the Bayesian impulse response. Figure 5 reports the system response of the model to an estimated supply shock. Following a productivity shock, physical capital is more productive, so households are more willing to postpone consumption and invest into physical capital to get more consumption in the future. Investment raises future productive capacity. Current output rises with productivity, while labour is fixed in this simple model and the capital used in current production is predetermined. The rise in income allows the households to increase immediately consumption. Note that the credible interval is computed across posterior parameter draws. It measures parameter uncertainty in the conditional impulse response, not unpredictable future shock realizations. VAR impulse-response intervals can also reflect parameter uncertainty.

2.4 Reloading results

Obtaining the posterior distribution can take quite a lot of time, that is why it is convenient if the model has not changed and if the posterior distribution is acceptable, then we can reload the results from the previous estimation as follows:

console response

```
estimation(datafile=mydata, first_obs=1,
mh_replic=0, prefilter=1, load_mh_file,
mode_file=RBC_estim_mode, mode_compute=0) y c k a;
```

Exercise 1: Fixing the estimation

1. If acceptance is very high and steps are tiny, explain how to adjust the proposal scale and which mixing diagnostics to inspect.
2. Estimate the capital intensity parameter α. Consult Smets and Wouters [2007] and document the exact parameterization and prior moments before transferring a prior.
3. Why must the steady state be updated?

3. A full-fledged RBC model

This model is inspired by the business cycle model of Hansen [1985] which aims at fixing anomalies in the real business cycle theory of Lucas [1977] and Kydland and Prescott [1982]. Before Hansen, the intratemporal substitution between consumption and leisure was unable to explain large aggregate fluctuations in hours worked/employment.

3.1 The household

Our economy is populated by identical and infinitely lived households distributed over a continuum $j \in [0, 1]$. The representative household utility function reads as follows:

$$\mathcal{U}\left(C_{jt}, H_{jt}\right) = \frac{{C_{jt}}^{1-\sigma_C}}{1-\sigma_C} - \varepsilon_t^H \chi \frac{{H_{jt}}^{1+\sigma_L}}{1+\sigma_L}$$

Utility is increasing in consumption C_{jt} and decreasing in labor H_{jt} where σ_C and σ_L are curvature parameters while ε_t^H is a shock to the disutility of labor supply. Households in this economy maximise the expected value of their utility:

$$\max_{\{C_{jt}, H_{jt}, B_{jt}, K_{jt}, I_{jt}\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau \mathcal{U}\left(C_{j,t+\tau}, H_{j,t+\tau}\right), \tag{11}$$

where β is the discount factor. Households face an intertemporal problem, they find the value of consumption C_{jt}, hours worked H_{jt} and bonds B_{jt} which delivers the highest value of utility. They face the following budget constraint which binds every period:

$$w_t H_{jt} + z_t K_{jt-1} + r_{t-1} B_{jt-1} = C_{jt} + I_{jt} + S\left(I_{jt}, I_{jt-1}\right) + B_{jt} + T_{jt}, \tag{12}$$

where w_t is the real wage, z_t the remuneration of capital services, r_{t-1} is the gross real return on bonds subscribed in $t-1$ and T_{jt} are lump-sum taxes used to finance the government budget. Here, $S\left(I_t, I_{t-1}\right)$

is an investment cost function, which replicates congestion costs when firms are simultaneously investing at the same time. The functional form reads as:

$$S\left(I_{jt}, I_{jt-1}\right) = \frac{\kappa}{2}\left(\frac{I_{jt}}{I_{jt-1}} - 1\right)^2 I_{jt-1} \tag{13}$$

where $\kappa \geq 0$ is an investment cost parameter.

The law of motion of capital also constrains the household's optimization problem:

$$K_{jt} = \varepsilon_t^I I_{jt} + (1-\delta) K_{jt-1}, \tag{14}$$

where K_{jt} is the stock of capital bought in t (the lag is explained by the time-to-build hypothesis regarding capital setting, *i.e.* physical capital requires one period to be installed), I_{jt} denotes investment in t while parameter δ denotes the depreciation rate of physical capital. The Lagrangian problem is given by:

$$\mathcal{L}_t = \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau \left[\begin{array}{c} \frac{C_{jt+\tau}^{1-\sigma_C}}{1-\sigma_C} - \varepsilon_{t+\tau}^H \chi \frac{{H_{jt+\tau}}^{1+\sigma_L}}{1+\sigma_L} \\ +\lambda_{t+\tau}^c \left[\begin{array}{c} w_{t+\tau} H_{jt+\tau} + z_{t+\tau} K_{jt-1+\tau} + r_{t-1+\tau} B_{jt-1+\tau} \\ -C_{jt+\tau} - I_{jt+\tau} - S\left(I_{jt+\tau}, I_{jt-1+\tau}\right) - B_{jt+\tau} - T_{jt+\tau} \end{array} \right] \\ +\lambda_{t+\tau}^k \left[\varepsilon_{t+\tau}^I I_{jt+\tau} + (1-\delta) K_{jt-1+\tau} - K_{jt+\tau}\right] \end{array} \right]$$

where λ_t^c and λ_t^k are Lagrange multipliers associated with the budget constraint and the physical capital accumulation constraint. The first order conditions are given by the welfare maximisation in C_{jt}, H_{jt}, I_{jt}, K_{jt} and B_{jt}:

$$\begin{aligned}
&(\partial C_{jt}) : \lambda_t^c = {C_{jt}}^{-\sigma_C} \\
&(\partial H_{jt}) : \varepsilon_t^H \chi H_{jt}^{\sigma_L} = \lambda_t^c w_t \\
&(\partial B_{jt}) : \lambda_t^c = \beta r_t \mathbb{E}_t\{\lambda_{t+1}^c\} \\
&(\partial I_{jt}) : -\lambda_t^c[1 + S_1(I_{jt}, I_{jt-1})] - \beta E_t[\lambda_{t+1}^c S_2(I_{jt+1}, I_{jt})] + \varepsilon_t^I \lambda_t^k = 0 \\
&(\partial K_{jt}) : \lambda_t^k = \beta E_t[(1-\delta)\lambda_{t+1}^k + \lambda_{t+1}^c z_{t+1}]
\end{aligned}$$

Substituting the Lagrange multipliers, our optimal conditions boil down to an Euler equation:

$$1 = \beta r_t \mathbb{E}_t \left\{ \left(\frac{C_{jt+1}}{C_{jt}} \right)^{-\sigma_C} \right\}, \tag{15}$$

a labour-supply equation:

$$w_t = \varepsilon_t^H \chi {H_{jt}}^{\sigma_L} {C_{jt}}^{\sigma_C} \tag{16}$$

It is possible to derive from the standard RBC model the endogenous asset price denoted $q_t = \lambda_t^k / \lambda_t^c$, defined as the ratio between the marginal cost of acquiring additional capital (Lagrangian multiplier on capital accumulation constraint λ_t^k) and the marginal utility of consumption (Lagrangian multiplier on budget constraint λ_t^c). Variable q_t also refers to the well known Tobin's Q published in Tobin [1969]. Tobin's Q is the ratio between a physical asset's market value and its replacement value, this ratio has considerable macroeconomic implications as it constitutes the standard bridge between financial markets and markets for goods and services. Here, constant steady-state investment and $\bar{\varepsilon}^I = 1$ imply $\bar{q} = 1$ through the investment condition. This value is model-specific, not a universal asset-pricing restriction.

For the specified cost function, the derivatives are

$$S_1(I, J) = \kappa(I/J - 1), \qquad S_2(I, J) = \frac{\kappa}{2}[1 - (I/J)^2].$$

Substituting $\lambda_t^k = \lambda_t^c q_t$ into the investment condition gives

$$\varepsilon_t^I q_t = 1 + S_1(I_{jt}, I_{jt-1}) + \beta E_t \left[\frac{\lambda_{t+1}^c}{\lambda_t^c} S_2(I_{jt+1}, I_{jt}) \right] \tag{17}$$

$$= 1 + \kappa \left(\frac{I_{jt}}{I_{jt-1}} - 1 \right) - \beta E_t \left[\frac{\lambda_{t+1}^c}{\lambda_t^c} \frac{\kappa}{2} \left(\left(\frac{I_{jt+1}}{I_{jt}} \right)^2 - 1 \right) \right]. \tag{18}$$

It is straightforward to see that q_t is the real shadow value of capital goods and a non-linear function of the (expected) growth rate of investment. In absence of investment costs $\kappa = 0$ (and of the investment shock), the real value of physical capital goods becomes one.

While the first order condition on physical capital reads as:

$$\lambda_t^c q_t = \beta \mathbb{E}_t \left\{ \left[(1-\delta) \lambda_{t+1}^c q_{t+1} + \lambda_{t+1}^c z_{t+1} \right] \right\} \tag{19}$$

$$q_t = \beta \mathbb{E}_t \left\{ \frac{\lambda_{t+1}^c}{\lambda_t^c} \left[(1-\delta) q_{t+1} + z_{t+1} \right] \right\} \tag{20}$$

$$q_t \simeq \frac{1}{r_t} \mathbb{E}_t \left\{ \left[(1-\delta) q_{t+1} + z_{t+1} \right] \right\} \tag{21}$$

The last line is a first-order certainty-equivalent expression around the deterministic steady state. The preceding line retains the stochastic discount factor and is the exact condition for risky capital. This is the no-arbitrage condition between risk-free assets (i.e. the real interest rate) and risky assets (physical capital). The no-arbitrage condition is determined by the equality between the current cost of purchasing one additional unit of physical capital (left side) and the expected discounted return of this unit. $1/r_t$ reduces the expected return and features the opportunity cost of risk free assets.

3.2 Firms

Firms are homogeneous and distributed on a unit interval $i \in [0, 1]$ and have the following production technology:

$$Y_{it} = \varepsilon_t^A K_{it-1}^{\alpha} H_{it}^{1-\alpha} \tag{22}$$

where Y_{it} is the production, K_{it-1} and H_{it} are input factors, i.e. the quantity of physical capital and hours involved in the production process of firms. Finally, ε_t^A denotes the productivity shock process.

The optimisation problem can be expressed in two ways.

The first one can be expressed as a profit maximisation problem:

$$\max_{Y_{it}, K_{it-1}, H_{it}} Y_{it} - z_t K_{it-1} - w_t H_{it} + \lambda_t \left[\varepsilon_t^A K_{it-1}^{\alpha} H_{it}^{1-\alpha} - Y_{it} \right]$$

where λ_t is the shadow value of the production constraint. Under perfect competition, the output first-order condition sets it equal to the unit output price.

The second way can be viewed as an inputs minimisation problem:

$$\min_{K_{it-1}, H_{it}} z_t K_{it-1} + w_t H_{it} - \lambda_t \left[\varepsilon_t^A K_{it-1}^{\alpha} H_{it}^{1-\alpha} - Y_{it} \right]$$

where λ_t denotes the Lagrange multiplier on the supply constraint and can also be interpreted as the marginal cost of producing an additional unit of good.

First order conditions are:

$$(\partial K_{it-1}) : z_t = \alpha \lambda_t \frac{Y_{it}}{K_{it-1}}$$
$$(\partial H_{it}) : w_t = (1-\alpha)\, \lambda_t \frac{Y_{it}}{H_{it}}$$
$$(\partial Y_{it}) : \lambda_t = 1$$

Replacing the Lagrange multiplier, the two conditions which determine the highest profits/lowest costs path are:

$$z_t = \alpha \frac{Y_{it}}{K_{it-1}} \tag{23}$$
$$w_t = (1-\alpha)\, \frac{Y_{it}}{H_{it}} \tag{24}$$

3.3 Aggregation and Equilibrium conditions

Our economy contains households j, firms i, and a government financed by lump-sum taxes. Their interactions, summarized by market clearing conditions for goods, labour and physical capital markets, can be expressed as follows:

$$\underbrace{\int_0^1 Y_{it} \mathrm{d}i}_{\text{aggregate supply}} = \underbrace{\int_0^1 C_{jt} \mathrm{d}j}_{\text{aggregate consumption}} + \underbrace{\int_0^1 I_{jt} + S\left(I_{jt}, I_{jt-1}\right) \mathrm{d}j}_{\text{aggregate investment}} + \underbrace{\bar{G}\varepsilon_t^G}_{\text{aggregate public spending}}$$

Public spending is driven by an exogenous demand process, with $\bar{G} = g^y \bar{Y}$, where g^y is the fixed spending-to-GDP ratio. Variable ε_t^G denotes the demand shock which affects output.

Letting Y_t, C_t and I_t denote the aggregate supply, consumption and investment, the resource constraint boils down to:

$$Y_t = C_t + I_t + S\left(I_t, I_{t-1}\right) + g^y \bar{Y} \varepsilon_t^G. \tag{25}$$

The equilibrium on the labour market is:

$$H_t = \underbrace{\int_0^1 H_{it} \mathrm{d}i}_{\text{labour demand}} = \underbrace{\int_0^1 H_{jt} \mathrm{d}j}_{\text{labour supply}}, \tag{26}$$

and for physical capital:

$$K_t = \underbrace{\int_0^1 K_{it} \mathrm{d}i}_{\text{capital demand}} = \underbrace{\int_0^1 K_{jt} \mathrm{d}j}_{\text{capital supply}}. \tag{27}$$

3.4 Disturbances

Our shock processes are characterized by four disturbances (in logs):

$$\log\left(\varepsilon_t^A\right) = \rho_A \log\left(\varepsilon_{t-1}^A\right) + \eta_t^A \quad \text{with } \eta_t^A \sim \mathcal{N}\left(0, \sigma_A^2\right), \tag{28}$$
$$\log\left(\varepsilon_t^G\right) = \rho_G \log\left(\varepsilon_{t-1}^G\right) + \eta_t^G \text{ with } \eta_t^G \sim \mathcal{N}\left(0, \sigma_G^2\right), \tag{29}$$
$$\log\left(\varepsilon_t^H\right) = \rho_H \log\left(\varepsilon_{t-1}^H\right) + \eta_t^H \text{ with } \eta_t^H \sim \mathcal{N}\left(0, \sigma_H^2\right), \tag{30}$$
$$\log\left(\varepsilon_t^I\right) = \rho_I \log\left(\varepsilon_{t-1}^I\right) + \eta_t^I \text{ with } \eta_t^I \sim \mathcal{N}\left(0, \sigma_I^2\right). \tag{31}$$

The innovations are mutually independent and independent over time. Shock levels are normalized to one at the deterministic steady state.

3.5 Summary of the model

The model contains 13 equations and 13 endogenous variables:

$$\{r_t, C_t, H_t, Y_t, K_t, I_t, q_t, z_t, w_t, \varepsilon_t^A, \varepsilon_t^G, \varepsilon_t^H, \varepsilon_t^I\}.$$

$$\begin{aligned}
1 &= \beta r_t E_t\left[(C_{t+1}/C_t)^{-\sigma_C}\right] \\
w_t &= \varepsilon_t^H \chi H_t^{\sigma_L} C_t^{\sigma_C} \\
q_t &= \beta E_t\left[(C_{t+1}/C_t)^{-\sigma_C}\left((1-\delta)q_{t+1} + z_{t+1}\right)\right] \\
\varepsilon_t^I q_t &= 1 + \kappa\left(I_t/I_{t-1} - 1\right) - \beta E_t\left[(C_{t+1}/C_t)^{-\sigma_C}\frac{\kappa}{2}\left((I_{t+1}/I_t)^2 - 1\right)\right] \\
K_t &= \varepsilon_t^I I_t + (1-\delta)K_{t-1} \\
Y_t &= \varepsilon_t^A K_{t-1}^{\alpha} H_t^{1-\alpha} \\
z_t &= \alpha\frac{Y_t}{K_{t-1}} \\
w_t &= (1-\alpha)\frac{Y_t}{H_t} \\
Y_t &= C_t + I_t + S\left(I_t, I_{t-1}\right) + g^y \bar{Y}\varepsilon_t^G \\
\log\left(\varepsilon_t^A\right) &= \rho_A \log\left(\varepsilon_{t-1}^A\right) + \eta_t^A \\
\log\left(\varepsilon_t^G\right) &= \rho_G \log\left(\varepsilon_{t-1}^G\right) + \eta_t^G \\
\log\left(\varepsilon_t^H\right) &= \rho_H \log\left(\varepsilon_{t-1}^H\right) + \eta_t^H \\
\log\left(\varepsilon_t^I\right) &= \rho_I \log\left(\varepsilon_{t-1}^I\right) + \eta_t^I
\end{aligned}$$

Steady state and normalization

With shock levels equal to one and no deterministic growth,

$$\bar{r} = \beta^{-1}, \quad \bar{q} = 1, \quad \bar{z} = \beta^{-1} - (1-\delta), \quad \bar{K} = \bar{H}(\bar{z}/\alpha)^{1/(\alpha-1)}.$$

Then $\bar{Y} = \bar{K}^{\alpha}\bar{H}^{1-\alpha}$, $\bar{I} = \delta\bar{K}$, $\bar{C} = (1-g^y)\bar{Y} - \bar{I}$ and $\bar{w} = (1-\alpha)\bar{Y}/\bar{H}$. Choose $\chi = \bar{w}/(\bar{H}^{\sigma_L}\bar{C}^{\sigma_C})$ to support the hours normalization, with $\bar{C} > 0$. Bonds are in zero net supply and the household optimum imposes no-Ponzi and transversality conditions. Production at t uses K_{t-1}, while K_t clears the end-of-period capital market and becomes productive next period. In fullRBC.mod, these steady-state coefficients are model-local expressions. They therefore update with α, β, δ and g^y rather than remaining fixed at the initial calibration.

3.6 Linearization

Define $\hat{x}_t = \log(X_t/\bar{X})$ and $\hat{\varepsilon}_t^j = \log\varepsilon_t^j$. For the gross bond return, $\hat{r}_t = \log(r_t/\bar{r})$, not a net-rate level. Adjustment costs vanish from resources at first order because S, S_1 and S_2 vanish at steady state. Their

derivatives still affect the investment condition. The linearized equations are:

$$
\begin{aligned}
\frac{1}{\sigma_C}\hat{r}_t &= \mathbb{E}_t\left\{\hat{c}_{t+1}\right\} - \hat{c}_t \\
\hat{w}_t &= \hat{\varepsilon}_t^H + \sigma_L \hat{h}_t + \sigma_C \hat{c}_t \\
\hat{q}_t &= \beta \mathbb{E}_t\left\{\left[(1-\delta)\hat{q}_{t+1} + \bar{z}\hat{z}_{t+1}\right]\right\} - \hat{r}_t \\
\hat{\varepsilon}_t^I + \hat{q}_t &= \kappa\left(\hat{\imath}_t - \hat{\imath}_{t-1}\right) - \beta\kappa E_t\left(\hat{\imath}_{t+1} - \hat{\imath}_t\right) \\
\hat{k}_t &= \delta\left(\hat{\varepsilon}_t^I + \hat{\imath}_t\right) + (1-\delta)\,\hat{k}_{t-1} \\
\hat{y}_t &= \hat{\varepsilon}_t^A + \alpha\hat{k}_{t-1} + (1-\alpha)\,\hat{h}_t \\
\hat{z}_t &= \hat{y}_t - \hat{k}_{t-1} \\
\hat{w}_t &= \hat{y}_t - \hat{h}_t \\
\hat{y}_t &= \frac{\bar{C}}{\bar{Y}}\hat{c}_t + \frac{\bar{I}}{\bar{Y}}\hat{\imath}_t + g^y\hat{\varepsilon}_t^G \\
\hat{\varepsilon}_t^A &= \rho_A\hat{\varepsilon}_{t-1}^A + \eta_t^A \\
\hat{\varepsilon}_t^G &= \rho_G\hat{\varepsilon}_{t-1}^G + \eta_t^G \\
\hat{\varepsilon}_t^H &= \rho_H\hat{\varepsilon}_{t-1}^H + \eta_t^H \\
\hat{\varepsilon}_t^I &= \rho_I\hat{\varepsilon}_{t-1}^I + \eta_t^I
\end{aligned}
$$

It is also common to estimate DSGE models using observables expressed in growth rates. Therefore, letting ΔY_t denote the growth rate of output, the corresponding measurement equation reads as follows:

$$
\begin{aligned}
\Delta Y_t &= \log(Y_t/Y_{t-1}) \\
&= \log(Y_t) - \log(Y_{t-1}) + \log(\bar{Y}/\bar{Y}) \\
&= \log(Y_t/\bar{Y}) - \log(Y_{t-1}/\bar{Y}) \\
&\simeq \hat{y}_t - \hat{y}_{t-1}
\end{aligned}
$$

This measurement equation applies to stationary model output. To use raw GDP growth with a nonzero mean, add a trend-growth intercept or demean the observed growth series consistently. We initially estimate this model using output growth as the only observable. The expression above is in decimal log units. The supplied data and the measurement equation in fullRBC.mod use $100\Delta Y_t$, in percentage points. Use the same scaling for consumption and investment growth. The data script aligns all seven downloaded series and verifies consecutive quarters before taking differences. An internal gap or duplicate quarter stops execution rather than silently changing the growth horizon.

Exercise 2: Estimating the RBC model

Consider the code fullRBC.mod of the full-fledged RBC model.

1. Let us first include investment as an observable variable:
   (a) Add a measurement equation in the model's code for the growth of investment di_t.
   (b) Add a steady state for this variable.
   (c) Add this observable to the varobs.
   (d) We need to include another shock in the estimation. Which one should be picked? Usually, we should include the shock that is more likely to generate fluctuations on this variable. Here, the investment shock is very likely to provide strong fluctuations on investment, so we are going to include ρ_i and $\text{sd}(\eta_{it})$ as parameters to estimate.
   (e) Run the estimation.
   (f) Investment adjustment cost κ may be estimated. Use the priors of Smets and Wouters [2007]
2. Examine the historical decomposition of shock for the observable variables.
3. Examine the forecasting error decomposition.
4. Let us now include consumption as observable. Same as for investment, include dc_t in the model's block, steady state, and set it in varobs. Use the government spending shock to estimate this series.
5. Finally, let us now include hours worked.
   (a) The exercise treats the hours proxy as stationary around a constant mean. Check this assumption rather than imposing a first difference automatically. So we need to make our equations in Dynare consistent with the transformation of the data. Since the data are demeaned, the observable is the demeaned fraction of time per day devoted to work (the remaining part is the leisure), expressed in percentage points. The supplied series is manufacturing weekly hours, converted to a fraction of calendar time. It is an illustrative proxy, not an economy-wide measure. Document this mapping and the steady-state normalization. Include and linearize the observable:
   $$dh_t = 100(H_t - \bar{H})$$
   (b) Add the steady state of this variable.
   (c) Add this variable to the observable block.
   (d) Then estimate the remaining shock on the labor supply.
6. Some other structural parameters such as σ_C and σ_L may be estimated. Use the priors of Smets and Wouters [2007].

Exercise 3: External habits

Let us assume external consumption habits:

$$U\left(C_{jt} - hC_{t-1}, H_{jt}\right) = \frac{\left(C_{jt} - hC_{t-1}\right)^{1-\sigma_C}}{1-\sigma_C} - \varepsilon_t^H \chi \frac{{H_{jt}}^{1+\sigma_L}}{1+\sigma_L}$$

Note that C_{t-1} is the previous average consumption in the population, while $C_{j,t-1}$ is agent specific. Habit Formation is based on catching up with the Joneses.
The household treats lagged aggregate consumption as external. Thus $\lambda_t^c = (C_{jt} - hC_{t-1})^{-\sigma_C}$, with $0 \leq h < 1$ and positive consumption surplus. There is no internal-habit derivative from future utility. The new marginal utility must enter all conditions containing λ_t^c, including investment and capital pricing. In particular:

$$\begin{aligned}
(\partial C_{jt}) &: \lambda_t^c = U'_{c,t} \\
(\partial H_{jt}) &: \varepsilon_t^H \chi H_{jt}^{\sigma_L} = \lambda_t^c w_t \\
(\partial B_{jt}) &: \lambda_t^c = \beta r_t \mathbb{E}_t\{\lambda_{t+1}^c\}
\end{aligned}$$

1. Update the first order conditions.
2. Find the steady state.
3. Linearize the equations.
4. Update the Dynare file and verify that $h = 0$ recovers the baseline.

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